

INDIAN INSTITUTE OF TECHNOLOGY DELHI

New Delhi – 110016

# INTERNSHIP REPORT

## Arduino Based Embedded Systems

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Entry No.:

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**Submitted To**

Guide Name

Designation

Organization

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# Chapter 1

## Introduction

This internship focused on the design and implementation of embedded systems using the Arduino UNO platform. The experiments cover basic digital interfacing, sensor integration, display modules, automation, and embedded applications.

The objective of the internship was to strengthen practical knowledge of microcontrollers, embedded programming, hardware interfacing, and real-time control systems through hands-on projects.

The report documents each experiment with its objective, circuit design, implementation methodology, working principle, results, and conclusion.

# Chapter 2

## Basic Arduino Experiments

### 2.1 LED Flash System

#### 2.1.1 Objective

The objective of this experiment is to interface an LED with the Arduino UNO and make it blink continuously at a fixed interval.

#### 2.1.2 Problem Statement

Design and implement an LED flashing system using Arduino UNO to understand digital output control and timing functions.

#### 2.1.3 Components Used

#### 2.1.4 Circuit Diagram

#### 2.1.5 Hardware Connections

#### 2.1.6 Software Used

- Arduino IDE
- SimulIDE

#### 2.1.7 Source Code

The complete Arduino implementation is available in the project repository.

[Click here to view the source code](#)

**2.1.8 Working Principle**

Write the explanation here.

**2.1.9 Results**

Write observations here.

**2.1.10 Conclusion**

Write conclusion here.